

University of Dhaka

Department of Computer Science and Engineering

Course: CSE 2213 (Data and Telecommunication Lab)

Project Name: Design and Implementation of an Amplitude Modulator and Demodulator Circuit with PCB Fabrication

Group No: B-11 (even)

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Introduction:

Amplitude modulation (AM) is one of the fundamental techniques in signal processing and communication systems. It involves varying the amplitude of a carrier wave in accordance with the amplitude of the modulating signal, which carries the desired information. Amplitude demodulation, on the other hand, is the process of extracting the original modulating signal from the modulated carrier wave. These techniques are widely used in various applications such as radio broadcasting, telemetry, and wireless communication systems.

This project focuses on the design and implementation of both amplitude modulator and demodulator circuits. The objective is to develop functional circuits on a neatly designed Printed Circuit Board (PCB) that perform these operations efficiently. The modulator circuit will accept two input signals – a carrier signal and a modulating signal – and generate an amplitude-modulated output. The demodulator circuit will then process the AM signal and recover the original modulating signal.

Why do We Need Modulation?

Size of the Antenna:

The size of an efficient antenna is proportional to the wavelength (λ) of the transmitted signal:

$$\lambda = c/f$$

where:

c = speed of light (3×10^8 m/s), f = frequency of the signal.

For a 300 Hz signal:

$$\lambda = (3 \times 10^8) / 300 = 1000 \text{ Km}$$

Such a large antenna is impractical.

Using a 596 kHz carrier wave:

$$\lambda = (3 \times 10^8) / 596000 = 503.36 \text{ m}$$

This significantly reduces the antenna size, making it feasible for communication systems.

Thus, modulation allows the use of higher carrier frequencies, enabling practical antenna sizes and efficient transmission.

Interference Reduction:

This is a point from the practical side of things. Suppose you transmit the baseband signal to a receiver, say, your friend's phone. Just like you, there will be thousands of people in the city using their mobile phones. There is no way to tell such signals apart, and they will interfere with each other leading to a lot of noise in the system and a very bad output. By using a carrier wave of high frequencies and allotting a band of frequencies to each message, there is no mixing up of signals, and the received signals are absolutely perfect.

Wireless Communication System:

Modulation has removed the necessity for using wires in the communication systems. It is because modulation is widely used in transmitting signals from one location to another with faster speed. Thus, the modulation technique has helped in enhancing wireless communication systems.

Components:

For Amplitude Modulator:

- Diode: 1N4148 (1 piece)
- Resistors:
 - 2.2 k Ω (2 pieces)
 - 10 k Ω (2 pieces)
- Inductor: 10 μ H (1 piece)
- Capacitor: 0.01 μ F (1 piece)

For Amplitude Demodulator:

- Diode: 1N34A (1 piece)
- Resistor: 10 k Ω (1 piece)
- Capacitor: 0.1 μ F (1 piece)

Additional Components:

- **PCB Board:** For permanent assembly.
- **Breadboard:** For initial prototyping and testing.
- **Connecting Wires:** For connections.
- **Probes:** For signal measurement.

Hardware:

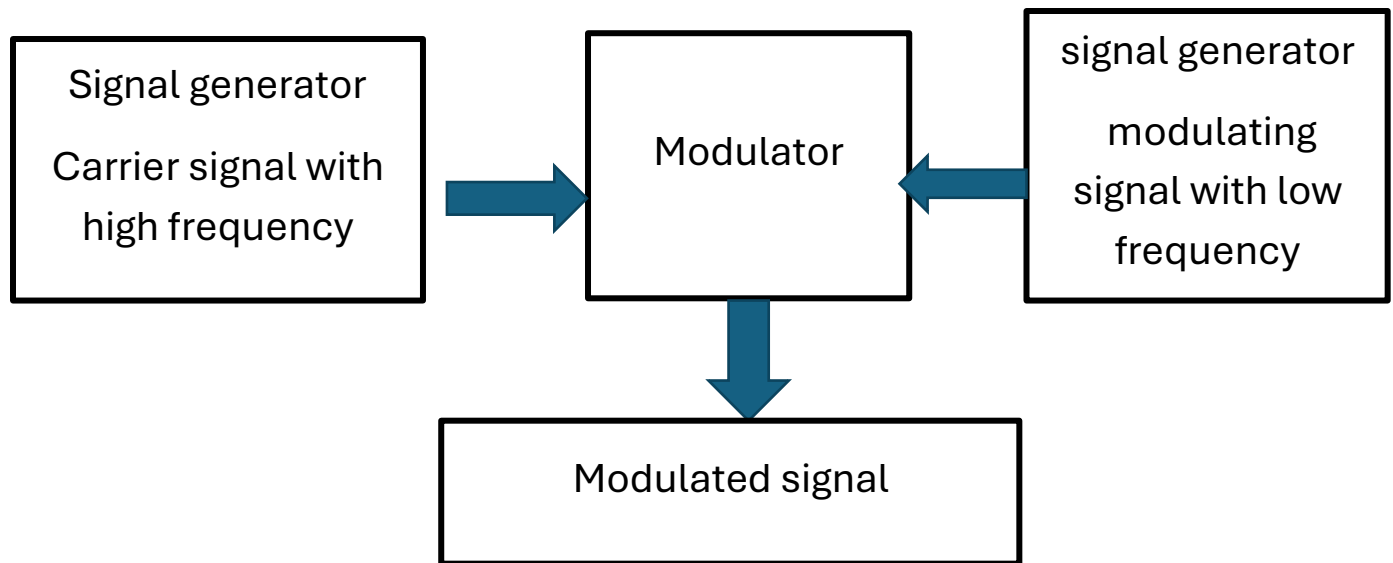
- **Signal Generator:** To produce the carrier and modulating signals.
- **Oscilloscope:** To visualize the modulated and demodulated signals.
- **Multimeter:** For measuring resistance and to check whether the circuit connection is shorted or not.
- **Soldering Kit:** For assembling the components onto the PCB.

Modulation:

Modulation is the process of varying a carrier signal to encode information from a modulating signal, enabling it to be transmitted over long distances or through different channels.

- **Carrier Signal:** This is a high-frequency sinusoidal wave, typically a continuous, unmodulated wave. It serves as the medium for transmitting the modulating signal.
- **Modulating Signal:** This is the information signal, such as audio or video, that contains the data to be transmitted. It modulates the carrier signal.
- **Modulated Signal:** This is the resultant waveform after the modulation process. The characteristics of the carrier signal (amplitude, frequency, or phase) are altered according to the modulating signal.
- **Principle:** In Amplitude Modulation (AM), for example, the amplitude of the carrier signal is varied proportionally to the instantaneous amplitude of the modulating signal, conveying the data encoded in the variations of the carrier's amplitude.

Block Diagram:



Mathematical Expressions:

1. Carrier Signal:

The carrier wave is a high-frequency sine wave used for modulation:

$$C(t) = A_c \sin (2\pi f_c t)$$

Where:

A_c : Amplitude of the carrier signal.

f_c : Frequency of the carrier signal.

t : Time.

2. Modulating Signal / Data Signal:

The modulating signal is the information signal:

$$m(t) = A_m \sin (2\pi f_m t)$$

A_m : Amplitude of the modulating signal.

f_m : Frequency of the modulating signal.

3. Amplitude Modulated Signal:

The amplitude of the carrier is varied based on the modulating signal:

$$AM(t) = [A_c + m(t)] \sin(2\pi f_c t)$$

Substitute $m(t)$,

$$AM(t) = A_c \sin(2\pi f_c t) + A_m \sin(2\pi f_m t) \sin(2\pi f_m t)$$

We get ,

$$AM(t) = A_c \sin(2\pi f_c t) + (A_m/2) [\cos(2\pi(f_c - f_m)t) - \cos(2\pi(f_c + f_m)t)]$$

4. Components of the AM Signal :

Carrier component

$$A_c \sin(2\pi f_c t)$$

Upper sideband

$$(A_m/2) \cos(2\pi(f_c + f_m)t)$$

Lower sideband

$$(A_m/2) \cos(2\pi(f_c - f_m)t)$$

5. Modulation Index:

The modulation index (μ) represents the extent of modulation:

$$\mu = A_m / A_c$$

$\mu < 1$: under modulation

$\mu = 1$: perfect modulation

$\mu > 1$: over modulation (causes distortion)

6. Bandwidth of AM :

$$\text{Bandwidth (BW)} = 2f_m$$

Where f_m is the frequency of the modulating signal.

7. Resonant Frequency in an AM Modulator Circuit:

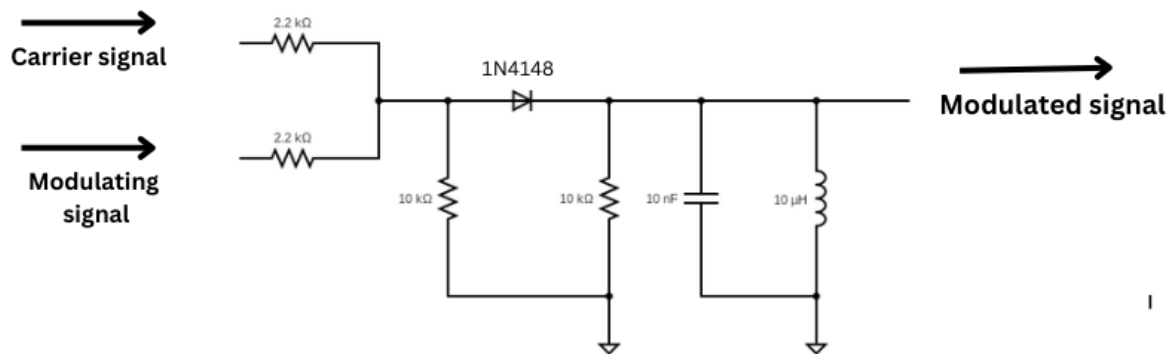
The resonant frequency (f) in the circuit diagram is calculated using the formula:

$$f = 1 / (2\pi\sqrt{LC})$$

The frequency is set by the inductor (L) and capacitor (C) values in the LC tank circuit, determining the carrier frequency for amplitude modulation (AM). It ensures the circuit filters and transmits the modulated signal effectively within the desired frequency range.

Amplitude Modulation Process and Component Explanation:

Modulator Circuit Diagram:



Input Signals: The carrier wave and message signal are introduced to the circuit. These signals are generated by signal generator. There are two resistors (2.2 kΩ) used with the input signals.

Resistors (2.2 kΩ) Purpose: Biasing and controlling current flow. These resistors limit the current through the diode and prevent circuit damage. They also help in maintaining proper signal levels.

Diode: 1N4148:

The diode rectifies the signals and creates a non-linear mixing of the carrier and modulating signals. The 1N4148 is a fast-switching diode, widely used in low-power signal applications. It rectifies the input signal, combining the modulating signal and the carrier wave to produce the amplitude-modulated signal.

Resistors (10 kΩ) Purpose:

Used as a load resistor and for proper voltage division. These resistors help shape the output waveform and maintain stability in the circuit.

Inductor and Capacitor Filtering:

Inductor creates a resonant circuit with the capacitor to filter the carrier wave. The inductor works with the capacitor to create a low-pass filter, ensuring that only the desired frequency

components are present in the output. Capacitor works with the inductor to smooth and filter the signal. It provides impedance at high frequencies and blocks DC components. This helps refine the modulated signal, allowing only the desired frequency to pass through.

Output

The amplitude-modulated signal is produced, ready for transmission or further processing.

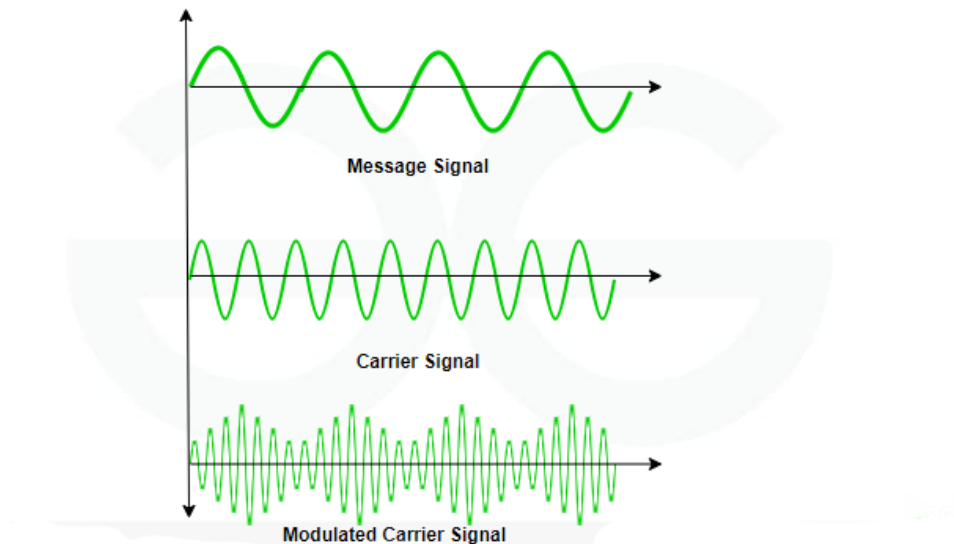
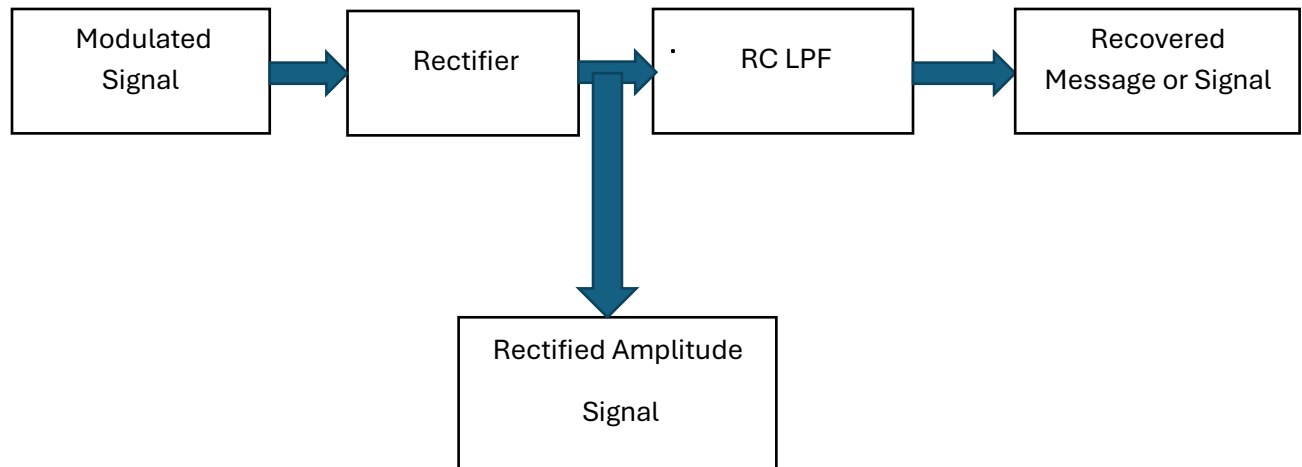


Fig: Modulation process

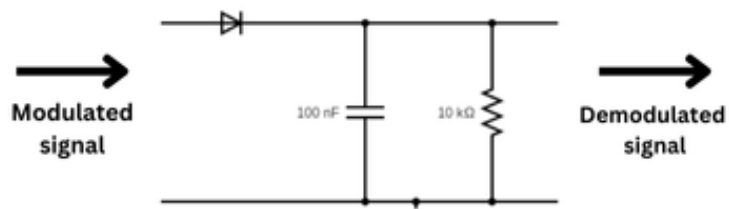
Demodulation:

Block Diagram:



Amplitude Demodulation Process and Component Explanation:

Demodulator Circuit Diagram:



Input AM Signal:

An AM signal consists of a high-frequency carrier modulated by a lower-frequency information signal.

Diode (1N34A):

Here diode acts as a rectifier in the demodulation process. It allows only one half (either positive or negative) of the incoming AM wave to pass through, effectively removing one side of the waveform. The 1N34A is a germanium diode, which has a low forward voltage drop (approximately 0.2-0.3V). This makes it highly sensitive and efficient for detecting weak AM signals, as it can rectify even low-amplitude signals without significant distortion.

Resistor (10 kΩ):

The resistor works with the capacitor to form an RC time constant that determines how quickly the circuit responds to changes in the signal. This value ensures that the resistor does not load the circuit heavily and allows for the proper discharge of the capacitor, maintaining the integrity of the recovered signal.

Capacitor (0.1 μF):

The capacitor acts as a filter, smoothing out the high-frequency carrier signal. It allows the lower-frequency baseband signal (the original information signal) to pass through while blocking the higher-frequency components. This capacitance value is chosen to effectively filter out the high-frequency components of the AM wave without distorting the baseband signal.

Output Signal:

At the output, we get a low-frequency signal that corresponds to the original information signal, with the carrier effectively removed.

Signals displayed on an oscilloscope:

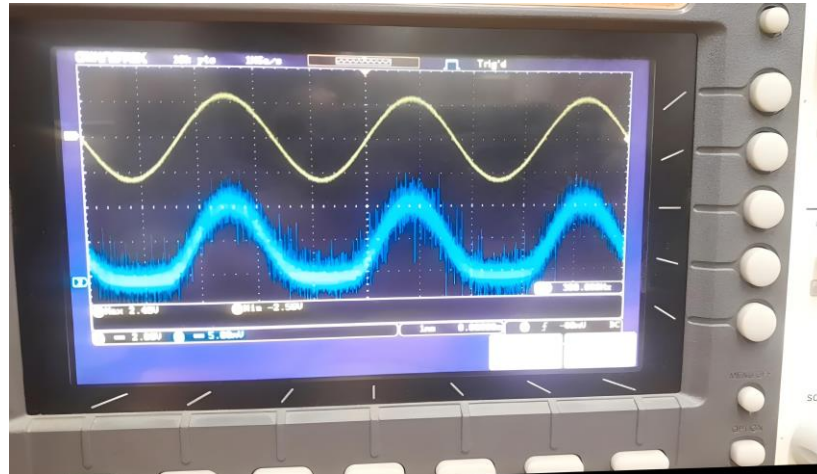
Input signal vs output signal:

- Video: https://drive.google.com/file/d/1Q7w_t-QiJiMwmqJhZssPhDs7423a6_ID/view?usp=sharing
- The **yellow waveform (upper trace)** is the **input modulating signal**, which is a low-frequency sinusoidal wave.
- The **blue waveform (lower trace)** is the **output demodulated signal**, which follows the shape of the input signal but has some high-frequency noise (possibly remnants of the carrier wave).

Observations:

- **Same Frequency:**
Both signals have the same fundamental frequency (300 Hz), meaning the demodulation circuit is correctly extracting the original modulating signal from the AM wave.
- **Noise in the Demodulated Signal:**
The blue waveform appears to have some **high-frequency ripples**, likely from the **remnants of the carrier wave**. This suggests that the low-pass filtering is not perfect, and some high-frequency components remain in the output.
This could be improved by using a capacitor with a slightly higher value in the demodulation circuit to enhance smoothing.
- **Demodulation Success:**

Despite the noise, the overall shape of the demodulated signal matches the input modulating signal, confirming that the circuit is successfully extracting the original information from the AM wave.



Modulated signal vs Demodulated signal:

- **Video:** https://drive.google.com/file/d/1Q5zjZPhg07SGqO-p_WmmHdSb_NL59qV9/view?usp=sharing
- **Yellow Waveform (Upper Trace) Modulated AM Signal:**
This waveform represents the **AM signal**, which consists of a high-frequency **carrier wave** modulated by a lower-frequency **message signal**.
- **Blue Waveform (Lower Trace) Demodulated Signal:**
This is the **demodulated output**, representing the extracted original message signal. The waveform closely resembles the **envelope** of the AM signal, meaning the demodulator is successfully recovering the original information. Some **high-frequency noise** is still present, likely residual components from the carrier that haven't been completely filtered out.

Observations:

- **Correct Demodulation:**

The demodulated signal retains the shape of the original modulating waveform. The **frequencies of both waveforms are the same**, confirming that the demodulation process successfully extracted the original signal from the AM carrier.

- **Residual Noise in the Demodulated Signal:**

The demodulated waveform still contains small **high-frequency ripples**, possibly due to an **insufficiently strong low-pass filter** in the demodulation circuit. A **slightly larger capacitor** (e.g., increasing from **0.1 μF to 0.22 μF**) could help in smoothing the signal further.



Waveform Amplitudes:

- a. **Max Voltage (V_max):** 248 mV
- b. **Min Voltage (V_min):** -272 mV
- c. These values indicate the peak-to-peak (V_pp) amplitude of the modulated signal.

Modulation Index Formula:

$$\mu = (V_{\text{max}} - V_{\text{min}}) / (V_{\text{max}} + |V_{\text{min}}|) \times 100$$

Plugging in the Values:

$$\mu = (248 - (-272)) / (248 + 272) \times 100$$

$$\mu = (248 + 272) / (248 + 272) \times 100$$

$$\mu = 520 / 520 \times 100 = 100\%$$

- **Potential Overmodulation Signs:**

If $v_{\min}$ reaches or exceeds $V_{\max}$ in magnitude, the carrier could be **cut off completely**, causing **overmodulation and distortion**.

- Looking at the lower portion of the waveform (blue trace), it seems **no phase reversal occurs**, suggesting **it is very close to 100% but not necessarily overmodulated**.

Problems Faced:

1. Prolonged Etching Process:

After designing the entire circuit on the PCB, we began the etching process. Initially, it took a long time to remove the copper from the PCB. However, after some time, we realized that agitating the etching solution (moving it back and forth) significantly sped up the reaction, reducing the overall time required.

2. Circuit Connection Error:

After completing the etching process and setting up all the components on the PCB, we encountered connection errors. Some components were not properly connected. Upon investigation, we discovered that the issue arose from the marker used during the circuit design on the PCB. We had forgotten to remove the marker after etching, and since the marker's ink acts as an insulator, it disrupted the circuit connections at some points.

To resolve this, we fixed the connections step by step using soldering. We manually inspected each connection line, and whenever we identified an error, we used solder to establish proper contact and fixed the issue.

3. Diode Problem:

After completing all the connections, we observed that while the modulated signal was displayed accurately, the demodulated signal was not. Upon troubleshooting, we identified the issue in the demodulation circuit. Replacing the diode in the demodulated circuit resolved the problem, and the proper output signal was successfully displayed.

4. Probe Problem:

We encountered issues with the probes, which prevented us from obtaining the desired signal on the oscilloscope. After addressing the problem, the correct signal was displayed.

Conclusion :

Amplitude modulation (AM) and demodulation are fundamental concepts in communication systems, playing a crucial role in transmitting and receiving information over long distances. Through AM, the information signal is encoded into the amplitude of a

carrier wave, making it suitable for efficient transmission. The demodulation process, on the other hand, recovers the original information signal from the modulated wave, enabling clear and accurate communication.

While amplitude modulation is simple and cost-effective, it is susceptible to noise and interference, which can impact signal quality. Proper demodulation techniques and circuit design, such as using high-quality diodes and effective filtering, ensure the accurate retrieval of the original signal. Together, modulation and demodulation remain integral to radio broadcasting, wireless communication, and numerous modern technologies.